

HW2 1D Gaussian Filtering with Wrap Condition

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1. 디바이스 정보

```
CUDA Device Query (Runtime API) version (CUDART static linking)

Detected 1 CUDA Capable device(s)

Device 0: "NVIDIA GeForce RTX 4080 SUPER"
  CUDA Driver Version / Runtime Version      13.2 / 13.2
  CUDA Capability Major/Minor version number: 8.9
  Total amount of global memory:             16376 MBytes (17170956288 bytes)
  (080) Multiprocessors, (128) CUDA Cores/MP: 10240 CUDA Cores
  GPU Max Clock rate:                        2595 MHz (2.60 GHz)
  Memory Clock rate:                         11501 Mhz
  Memory Bus Width:                          256-bit
  L2 Cache Size:                             67108864 bytes
  Maximum Texture Dimension Size (x,y,z)     1D=(131072), 2D=(131072, 65536), 3D=(16384, 16384, 16384)
  Maximum Layered 1D Texture Size, (num) layers 1D=(32768), 2048 layers
  Maximum Layered 2D Texture Size, (num) layers 2D=(32768, 32768), 2048 layers
  Total amount of constant memory:            65536 bytes
  Total amount of shared memory per block:    49152 bytes
  Total shared memory per multiprocessor:     102400 bytes
  Total number of registers available per block: 65536
  Warp size:                                 32
  Maximum number of threads per multiprocessor: 1536
  Maximum number of threads per block:        1024
  Max dimension size of a thread block (x,y,z): (1024, 1024, 64)
  Max dimension size of a grid size (x,y,z):  (2147483647, 65535, 65535)
  Maximum memory pitch:                       2147483647 bytes
  Texture alignment:                           512 bytes
  Concurrent copy and kernel execution:       Yes with 1 copy engine(s)
  Run time limit on kernels:                   Yes
  Integrated GPU sharing Host Memory:          No
  Support host page-locked memory mapping:     Yes
  Alignment requirement for Surfaces:          Yes
  Device has ECC support:                      Disabled
  CUDA Device Driver Mode (TCC or WDDM):       WDDM (Windows Display Driver Model)
  Device supports Unified Addressing (UVA):    Yes
  Device supports Managed Memory:              Yes
  Device supports Compute Preemption:          Yes
  Supports Cooperative Kernel Launch:          Yes
  Device PCI Domain ID / Bus ID / location ID: 0 / 1 / 0
  Compute Mode:
    < Default (multiple host threads can use ::cudaSetDevice() with device simultaneously) >

deviceQuery, CUDA Driver = CUDART, CUDA Driver Version = 13.2, CUDA Runtime Version = 13.2, NumDevs = 1
Result = PASS
```

Figure 1: nvidia-samples

```
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2025 NVIDIA Corporation
Built on Tue_Dec_16_19:27:18_Pacific_Standard_Time_2025
Cuda compilation tools, release 13.1, V13.1.115
Build cuda_13.1.r13.1/compiler.37061995_0
```

Figure 2: nvcc -version

Compute Capability 8.9, CUDA Toolkit 13.1 설치됨.

2. 실험

```

/*
Independent Variable:
_THREADS_PER_BLOCK 64, 128, 256, 512, 1024

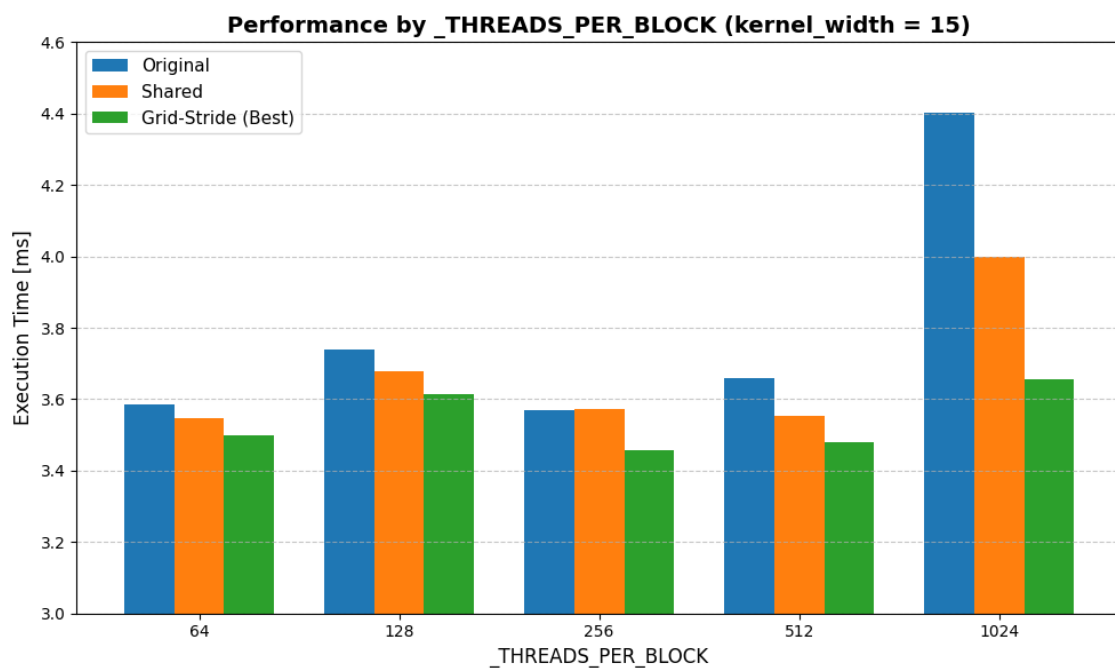
Controlled Variables:
_NUMBER_OF_TESTS_ON_GPU 32
_NUMBER_OF_ELEMENTS ((1 << 28) + 3)
_MIN_GRID_REDUCTION_FACTOR 2
_MAX_GRID_REDUCTION_FACTOR 32
_MIN_KERNEL_SIZE_DEVICE 1
_MAX_KERNEL_SIZE_DEVICE 15
*/

```

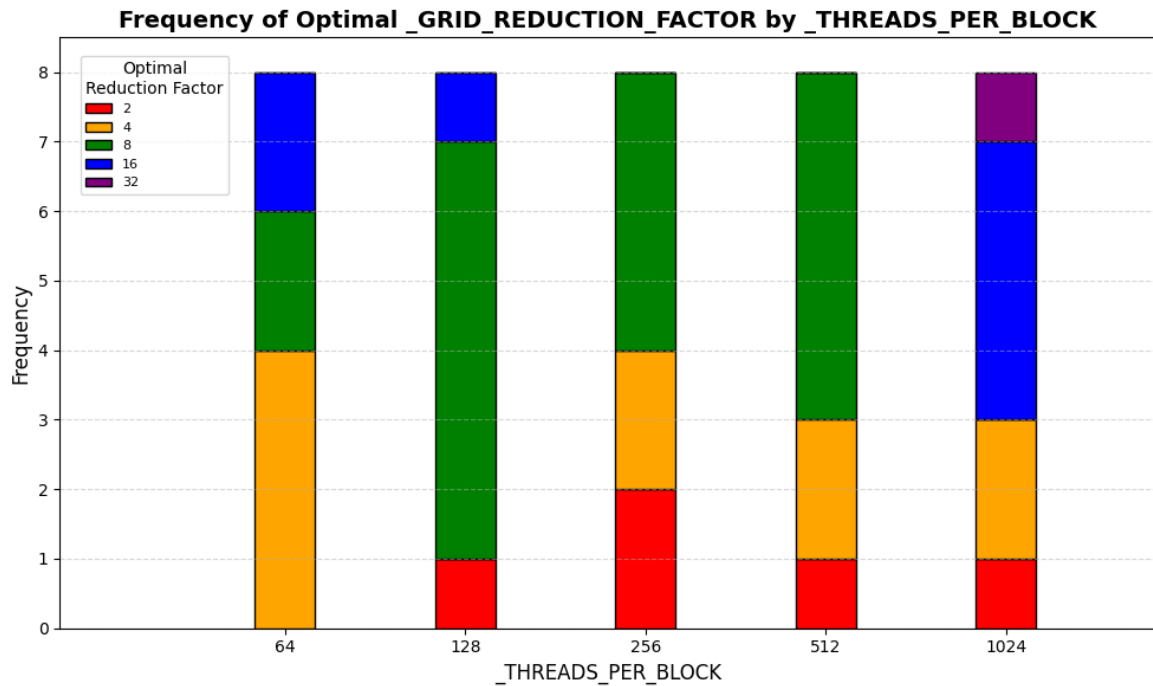
Figure 3: Variables

모든 실험은 `_NUMBER_OF_TESTS_ON_GPU = 32`인 상태에서 진행되었다. 즉, 모든 device에서의 측정 시간은 32회 측정의 평균이다. 더 선명한 결과를 얻기 위해 `_NUMBER_OF_ELEMENTS`는 기존 $((1 \ll 26) + 3)$ 에서 $((1 \ll 28) + 3)$ 으로 크기를 늘려서 측정했다.

2-1. `_THREADS_PER_BLOCK`의 변화에 따른 성능 차이



Graph 1: kernel_width 15, Execution Time by `_THREADS_PER_BLOCK`



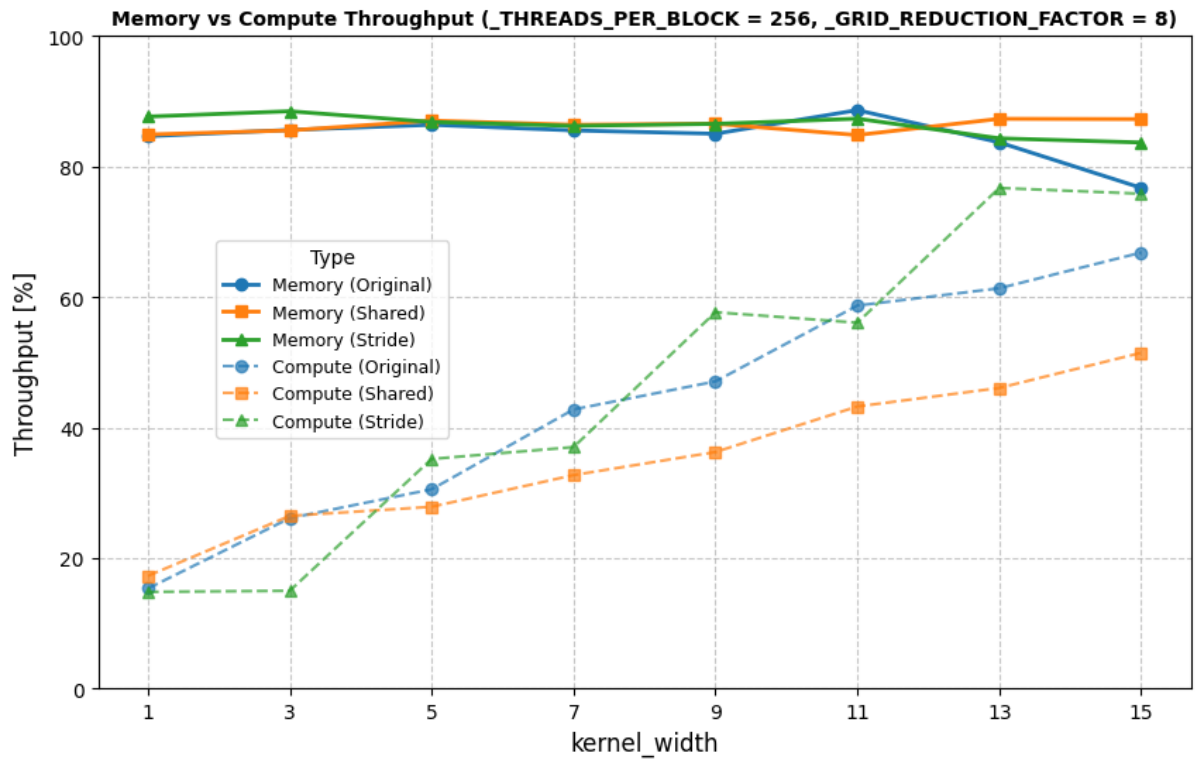
Graph 2: Optimal _GRID_REDUCTION_FACTOR by _THREADS_PER_BLOCK

바로 알 수 있는 것은, host는 kernel width가 증가할수록 연산 시간이 증가했지만 device의 경우 변화가 거의 없는 memory-bound 특성을 확인할 수 있다. 또한 kernel 함수의 3가지 종류 중 grid-stride loop이 가장 보편적으로 효과가 좋았음을 알 수 있다.

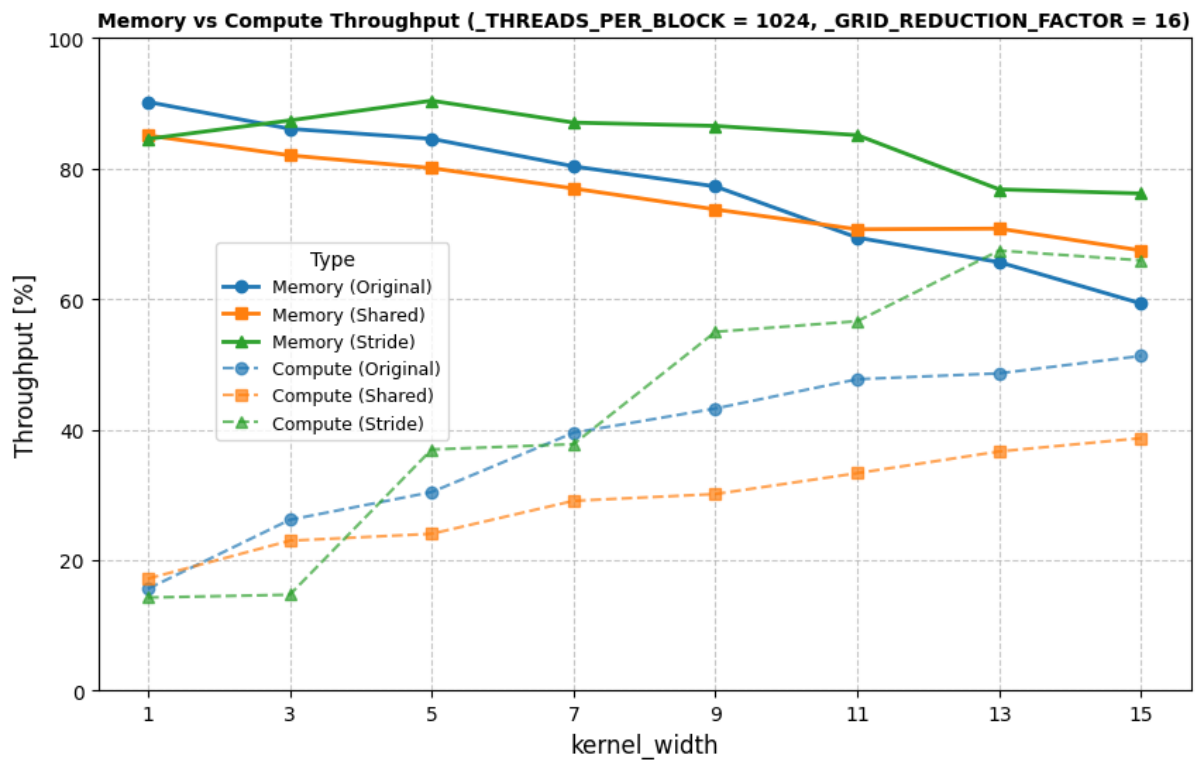
Shared memory의 경우 예전 GPU에서 확실하게 실행 시간의 단축을 불러올 수 있는 요소였는데, 이 실험에서는 그런 모습이 보이지 않았다. 이는 GPU의 발전으로 인해 L2 cache의 크기와 효율이 대폭 증가했기 때문으로 추측된다. 이는 nsight compute의 분석 결과를 보면 제대로 확인할 수 있을 것이다.

Grid-stride loop을 사용했을 때 일반적으로는 reduction factor가 4나 8일 때 가장 좋은 성능을 보인다. 예외로는 _THREADS_PER_BLOCK이 1024일 때였는데, 이때는 16이 가장 좋은 성능을 보였으나 절대적인 성능은 좋지 못하는데, 이는 block dimension이 너무 클 때 발생할 수 있는 문제로 역시 nsight compute로 warp occupancy를 확인하면 좋을 것 같다.

2-2. Nsight Compute 분석



Graph 3: _THREADS_PER_BLOCK 256, _GRID_REDUCTION_FACTOR 8, Throughput



Graph 4: _THREADS_PER_BLOCK 1024, _GRID_REDUCTION_FACTOR 16, Throughput

두 그래프는 각각 _THREADS_PER_BLOCK이 256, 1024일 때의 3가지 kernel의 memory와 compute throughput을 나타낸다. Grid-stride loop kernel의 reduction factor는 각각의 _THREADS_PER_BLOCK에서 최적이었던 빈도가 가장 높은 값으로 선정되었다.

두 그래프의 공통점으로 보면, grid-stride loop 방식에서 compute throughput이 정체되는 구간과 급증하는 구간이 반복적으로 나타난다는 것이다. 여기서 연산량의 증가와 도는 loop의 횟수 간의 관계를 짐작할 수 있다.

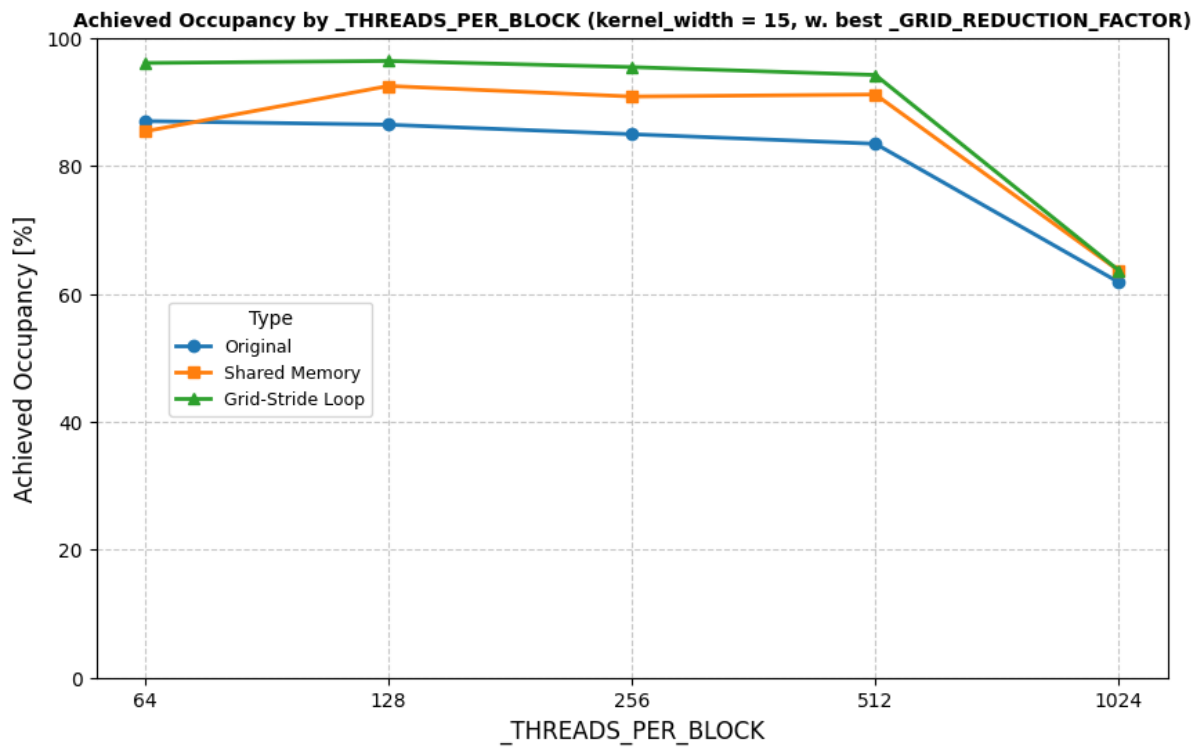
_THREADS_PER_BLOCK이 256인 경우, memory throughput은 일정하고 compute throughput은 증가하는 평이한 모습을 보인다. Shared memory 방식의 compute throughput이 original보다 낮게 나온 것은 original에서 필터 연산을 할 때마다 반복하는 if문이 전부 1번의 사전 작업으로 변환되었기 때문이다. 또한 grid-stride loop 방식의 compute throughput이 original보다 높게 나온 것은 연산 방식은 동일하지만 loop를 도느라 이중 for문을 사용하기 때문이다.

_THREADS_PER_BLOCK이 1024인 경우 memory throughput의 경향성은 동일하지만, memory throughput이 점차 감소한다. Original 방식은 block dimension이 너무 커져서 SM 안에 들어갈 수 있는 최대 thread 개수에 의해 warp occupancy가 떨어진 것으로 보인다(SM당 최대 1536개의 threads). Shared memory 방식은 shared memory를 사용했기 때문에 타격을 덜 입었고, grid-stride loop 방식은 original과 다르게 반복적으로 block을 재활용하기 때문에 block 교체에 드는 비용이 없다.

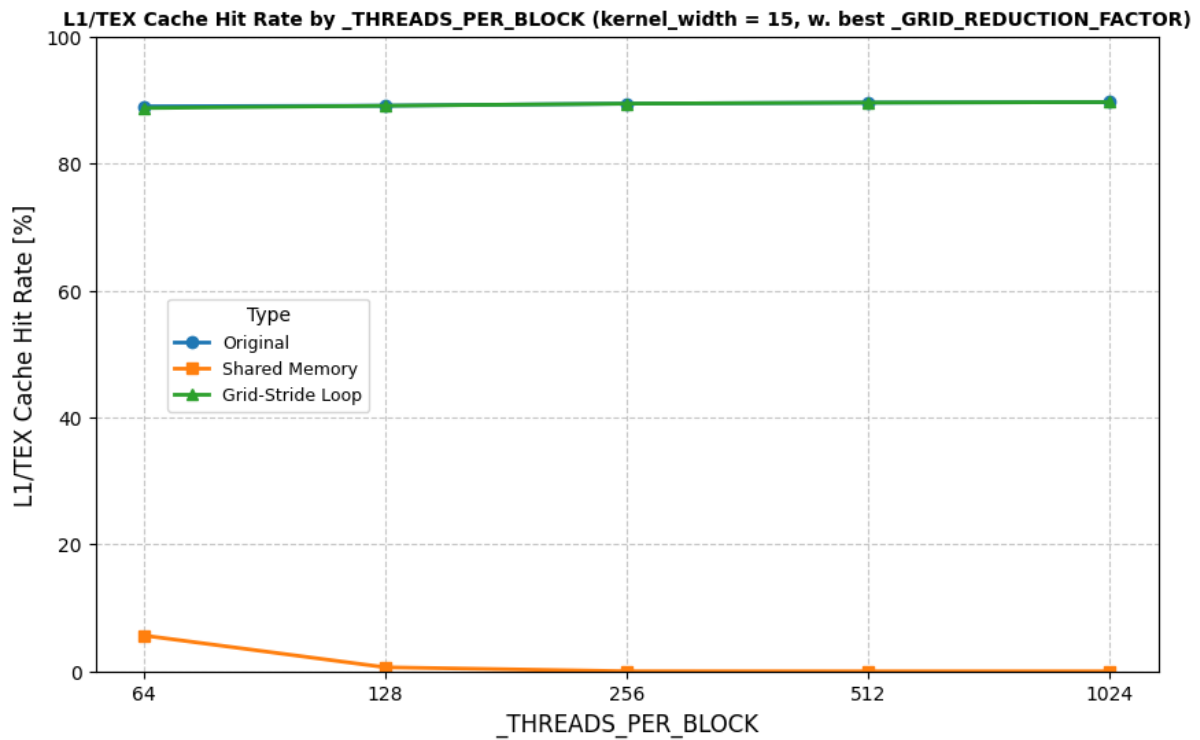
Kernel	Original	Shared Memory	Grid-Stride Loop
# of registers	23	22	28

Table 1: Number of Registers

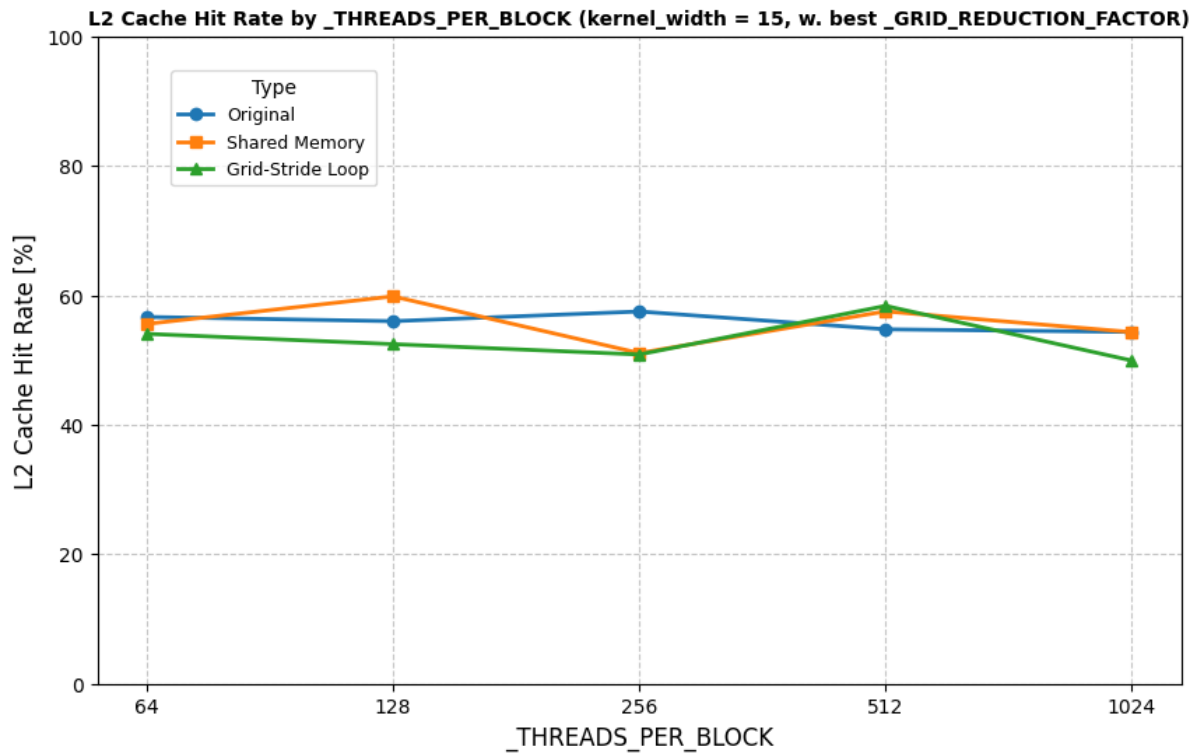
Block당 최대 register 개수는 65536개로 $28 \times 1024 = 28672$ 의 값은 한참 못 미친다. register에 의한 병목은 없는 것으로 판명됐다. 이제 다음으로 achieved occupancy와 L1/TEX, L2 cache hit rate, 그리고 shared memory 사용량을 볼 것이다.



Graph 5: Achieved Occupancy by _THREADS_PER_BLOCK



Graph 6: L1/TEX Cache Hit Rate by _THREADS_PER_BLOCK



Graph 7: L2 Cache Hit Rate by _THREADS_PER_BLOCK

_THREADS_PER_BLOCK	64	128	256	512	1024
Dynamic Shared Memory per Block [kbyte/block]	0.312	0.568	1.08	2.10	4.15
$4 * (14 + _THREADS_PER_BLOCK)$	312	568	1080	2104	4152

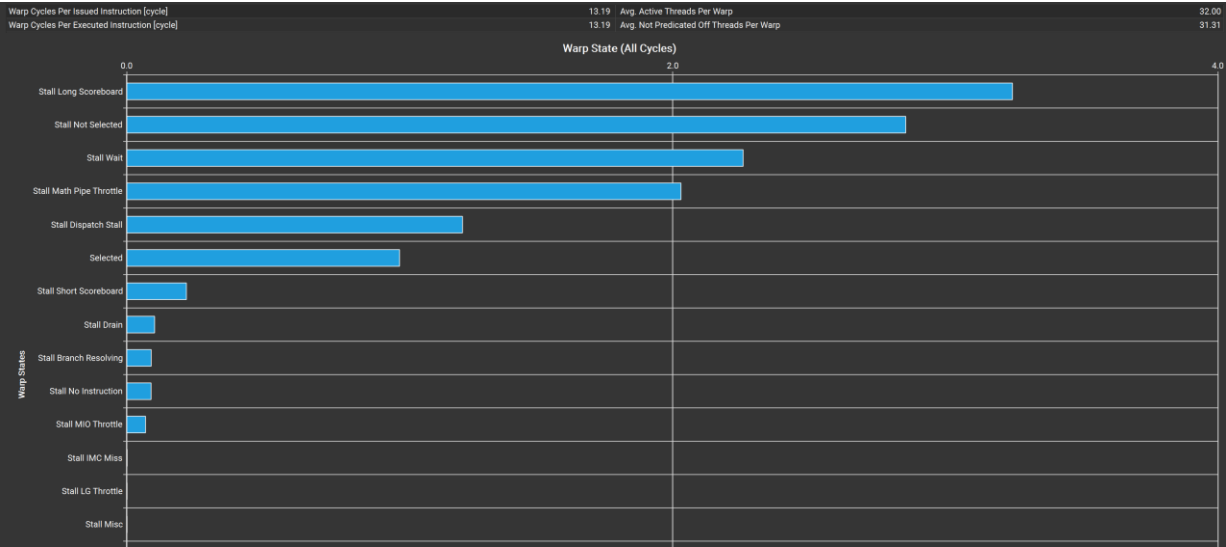
Table 2: Shared Memory per Block by _THREADS_PER_BLOCK

Achieved occupancy를 살펴보니 이전에 생각했던 내용이 대부분 맞아떨어진 것을 볼 수 있다. _THREADS_PER_BLOCK이 1024가 되기 전까지는 안정적이었다가, 과도한 block 크기 때문에 1개의 SM에 block이 하나밖에 들어가지 않으면서 achieved occupancy가 낮아질 것으로 예상했다. 하지만 3개의 kernel 모두 같이 occupancy가 낮아지는 것으로는 이전에 확인한 memory throughput의 양상을 설명하기 힘들다.

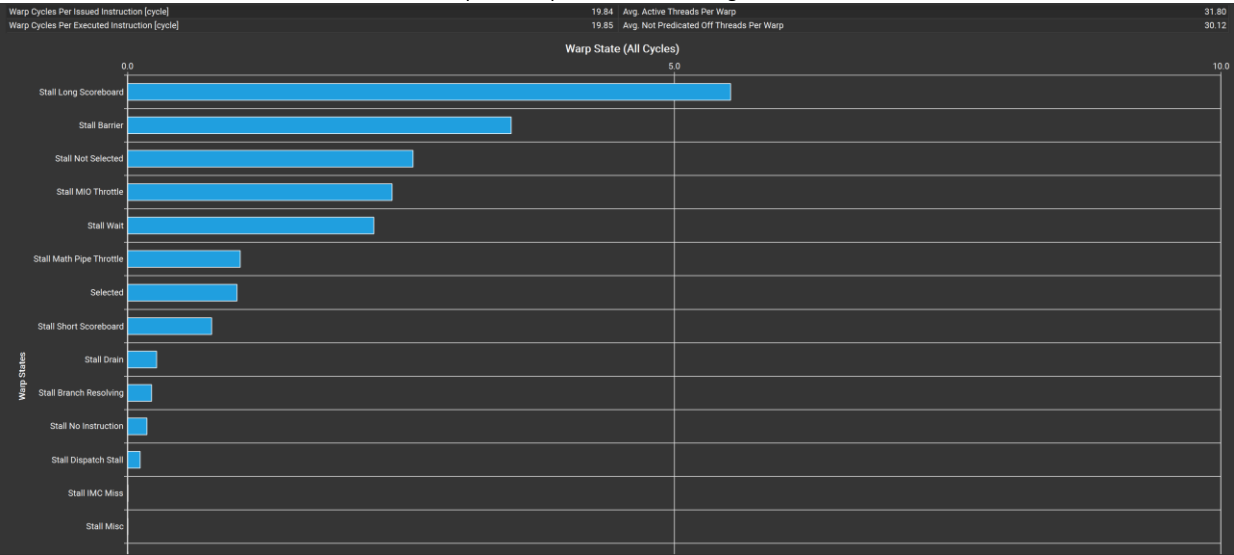
그래서 L1/TEX와 L2 cache의 hit rate를 살펴봤다. Original과 grid-stride loop의 경우 L1 cache의 hit rate가 90%에 가까운 값을 냈고, shared memory는 L2 cache를 거쳐서 shared memory로 데이터가 가기 때문에 L1 cache를 사용하지 않는다는 것이 제대로 나타나있다. L2 cache hit rate가 전부 비슷한 이유는 original과 grid-stride loop kernel은 filtering 연산을 하기 위해 중복 데이터를 접근하고, shared memory kernel의 경우 shared memory에 데이터를 넘기기 위해 L2 cache를 거쳐가기 때문이다. Shared memory kernel은 초기에 데이터를 shared memory에 적재한 후에 실제 연산을 전부 shared memory에서 처리하기 때문에 L2 cache에 접근하는 절대적인 횟수가 낮고, hit rate는 cache에 접근한 횟수를 기준으로 계산된다.

하지만 이것만으로 original kernel이 latency hiding에 실패했는지 명확하게 알 수 없다. 커널 간

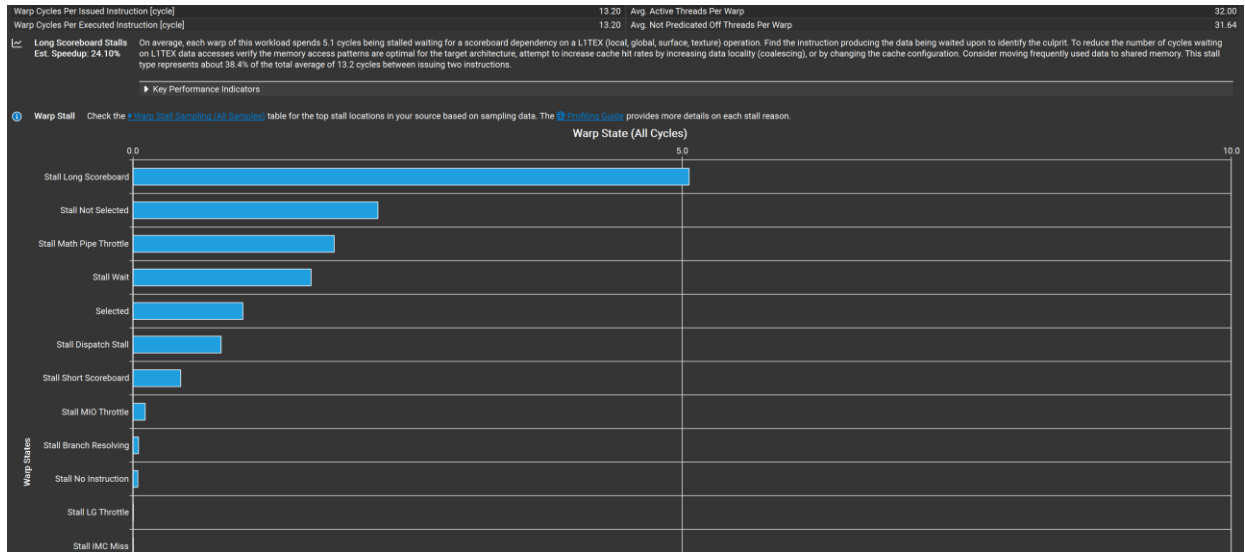
의 명확한 차이가 나는 원인을 알아보기 위해 full metric으로 다시 돌려서 warp state statistics 정보를 불러왔다.



Graph 8: Warp State Statistics, Original



Graph 9: Warp State Statistics, Shared Memory



Graph 10: Warp State Statistics, Grid-Stride Loop

Shared memory는 stall barrier, 즉 syncthread에 의한 동기화 과정에서 cycle을 낭비하는 것을 확인할 수 있었다. Original은 grid-stride loop과 비교하면 stall long scoreboard 이외의 나머지 지표가 훨씬 크게 보이고, 이것이 original kernel이 memory throughput이 급락하고 latency hiding이 잘 되지 않는 원인으로 볼 수 있다. Not selected나 wait 등 스케줄링 중에 생긴 overhead로 그 시간에 실제 연산을 할 수 없었다.

3. 표와 스크린샷

3-1-1. _THREADS_PER_BLOCK = 64

Kernel Width	Host [ms]	Original [ms] (Speedup)	Shared [ms] (Speedup)	Stride [ms] (Speedup)	Best Reduction Factor
1	247.461	3.676 (67.315x)	3.499 (70.714x)	3.430 (72.147x)	8
3	545.829	3.498 (156.056x)	3.572 (152.800x)	3.479 (156.882x)	8
5	498.344	3.565 (139.798x)	3.575 (139.381x)	3.533 (141.071x)	16
7	820.022	3.545 (231.301x)	3.556 (230.610x)	3.500 (234.284x)	4
9	798.199	3.534 (225.874x)	3.595 (222.030x)	3.483 (229.137x)	4
11	1090.869	3.561 (306.348x)	3.611 (302.069x)	3.511 (310.662x)	4
13	1096.011	3.527 (310.774x)	3.514 (311.874x)	3.595 (304.882x)	16
15	1384.846	3.587 (386.118x)	3.547 (390.463x)	3.499 (395.781x)	4

Table 3: _THREADS_PER_BLOCK 64, Time Comparison

Reduction Factor	2	4	8	16	32
Time (1) [ms]	3.486	3.738	3.430	3.686	3.559
Time (3) [ms]	3.722	3.540	3.479	3.630	3.732
Time (5) [ms]	3.613	3.547	3.674	3.533	3.652
Time (7) [ms]	3.616	3.500	3.534	3.820	3.810
Time (9) [ms]	3.647	3.483	3.738	3.803	3.930
Time (11) [ms]	3.673	3.511	3.712	3.784	3.708
Time (13) [ms]	3.735	3.795	3.742	3.595	3.653

Time (15) [ms]	3.527	3.499	3.612	3.757	3.823
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Table 4: _THREADS_PER_BLOCK 64, Grid-Stride Loop Comparison

3-1-2. _THREADS_PER_BLOCK = 128

Kernel Width	Host [ms]	Original [ms] (Speedup)	Shared [ms] (Speedup)	Stride [ms] (Speedup)	Best Reduction Factor
1	251.145	3.701 (67.864x)	3.504 (71.676x)	3.483 (72.101x)	8
3	540.318	3.552 (152.105x)	3.542 (152.530x)	3.494 (154.653x)	8
5	483.525	3.546 (136.343x)	3.577 (135.159x)	3.514 (137.619x)	8
7	788.396	3.588 (219.753x)	3.554 (221.804x)	3.531 (223.278x)	8
9	753.431	3.546 (212.498x)	3.596 (209.542x)	3.508 (214.760x)	8
11	1050.919	3.541 (296.774x)	3.583 (293.275x)	3.517 (298.816x)	8
13	1086.893	3.549 (306.277x)	3.577 (303.886x)	3.527 (308.146x)	2
15	1345.660	3.740 (359.798x)	3.679 (365.765x)	3.613 (372.487x)	16

Table 5: _THREADS_PER_BLOCK 128, Time Comparison

Reduction Factor	2	4	8	16	32
Time (1) [ms]	3.545	3.520	3.483	3.517	3.620
Time (3) [ms]	3.540	3.520	3.494	3.564	3.677
Time (5) [ms]	3.533	3.531	3.514	3.597	3.748
Time (7) [ms]	3.562	3.546	3.531	3.609	3.743
Time (9) [ms]	3.539	3.558	3.508	3.582	3.711
Time (11) [ms]	3.536	3.562	3.517	3.588	3.722
Time (13) [ms]	3.527	3.697	3.638	3.677	3.983
Time (15) [ms]	3.668	3.676	3.715	3.613	3.900

Table 6: _THREADS_PER_BLOCK 128, Grid-Stride Loop Comparison

3-1-3. _THREADS_PER_BLOCK = 256

Kernel Width	Host [ms]	Original [ms] (Speedup)	Shared [ms] (Speedup)	Stride [ms] (Speedup)	Best Reduction Factor
1	248.795	3.717 (66.934x)	3.517 (70.750x)	3.474 (71.609x)	8
3	542.945	3.589 (151.260x)	3.565 (152.303x)	3.495 (155.345x)	8
5	490.738	3.583 (136.971x)	3.564 (137.711x)	3.511 (139.789x)	8
7	789.818	3.561 (210.800x)	3.585 (220.332x)	3.513 (224.809x)	4
9	776.656	3.586 (216.559x)	3.584 (216.695x)	3.538 (219.502x)	4
11	1051.538	3.564 (295.009x)	3.587 (293.192x)	3.517 (298.973x)	8
13	1073.116	3.602 (297.933x)	3.555 (301.865x)	3.453 (310.809x)	2
15	1392.491	3.569 (390.155x)	3.573 (389.677x)	3.458 (402.643x)	2

Table 7: _THREADS_PER_BLOCK 256, Time Comparison

Reduction Factor	2	4	8	16	32
Time (1) [ms]	3.529	3.510	3.474	3.480	3.578
Time (3) [ms]	3.547	3.530	3.495	3.549	3.647
Time (5) [ms]	3.555	3.536	3.511	3.597	3.698
Time (7) [ms]	3.553	3.513	3.517	3.583	3.706

Time (9) [ms]	3.690	3.538	3.701	3.569	3.691
Time (11) [ms]	3.560	3.566	3.517	3.588	3.692
Time (13) [ms]	3.453	3.527	3.495	3.548	3.661
Time (15) [ms]	3.458	3.521	3.543	3.557	3.700

Table 8: _THREADS_PER_BLOCK 256, Grid-Stride Loop Comparison

3-1-4. _THREADS_PER_BLOCK = 512

Kernel Width	Host [ms]	Original [ms] (Speedup)	Shared [ms] (Speedup)	Stride [ms] (Speedup)	Best Reduction Factor
1	247.895	3.691 (67.157x)	3.568 (69.472x)	3.435 (72.159x)	4
3	539.583	3.563 (151.425x)	3.601 (149.842x)	3.484 (154.878x)	4
5	488.949	3.560 (137.349x)	3.602 (135.747x)	3.511 (139.243x)	8
7	793.217	3.598 (220.447x)	3.577 (221.726x)	3.511 (225.939x)	8
9	762.137	3.557 (214.270x)	3.590 (212.300x)	3.502 (217.611x)	8
11	1049.342	3.535 (296.878x)	3.578 (293.295x)	3.508 (299.102x)	8
13	1069.526	3.438 (311.110x)	3.582 (298.575x)	3.474 (307.824x)	2
15	1371.823	3.660 (374.862x)	3.553 (386.153x)	3.481 (394.068x)	8

Table 9: _THREADS_PER_BLOCK 512, Time Comparison

Reduction Factor	2	4	8	16	32
Time (1) [ms]	3.544	3.435	3.472	3.461	3.574
Time (3) [ms]	3.557	3.484	3.487	3.535	3.632
Time (5) [ms]	3.563	3.568	3.511	3.578	3.708
Time (7) [ms]	3.594	3.556	3.511	3.569	3.691
Time (9) [ms]	3.558	3.588	3.502	3.552	3.672
Time (11) [ms]	3.564	3.597	3.508	3.571	3.665
Time (13) [ms]	3.474	3.521	3.486	3.528	3.646
Time (15) [ms]	3.497	3.487	3.481	3.542	3.642

Table 10: _THREADS_PER_BLOCK 512, Grid-Stride Loop Comparison

3-1-5. _THREADS_PER_BLOCK = 1024

Kernel Width	Host [ms]	Original [ms] (Speedup)	Shared [ms] (Speedup)	Stride [ms] (Speedup)	Best Reduction Factor
1	250.542	3.547 (70.640x)	3.420 (73.248x)	3.473 (72.146x)	16
3	540.608	3.444 (156.959x)	3.530 (153.162x)	3.417 (158.199x)	2
5	492.841	3.558 (138.504x)	3.610 (136.535x)	3.444 (143.105x)	4
7	797.266	3.556 (224.235x)	3.680 (216.670x)	3.438 (231.892x)	4
9	766.937	3.634 (211.052x)	3.753 (204.361x)	3.422 (224.100x)	16
11	1071.491	3.841 (278.932x)	3.878 (276.288x)	3.421 (313.235x)	16
13	1113.641	3.977 (280.031x)	3.865 (288.153x)	3.668 (303.631x)	32
15	1363.692	4.401 (309.894x)	3.999 (340.972x)	3.655 (373.091x)	16

Table 11: _THREADS_PER_BLOCK 1024, Time Comparison

Reduction Factor	2	4	8	16	32
Time (1) [ms]	3.546	3.577	3.509	3.473	3.545

Time (3) [ms]	3.417	3.522	3.494	3.493	3.567
Time (5) [ms]	3.453	3.444	3.462	3.503	3.582
Time (7) [ms]	3.449	3.438	3.464	3.484	3.579
Time (9) [ms]	3.681	3.576	3.446	3.422	3.507
Time (11) [ms]	3.662	3.577	3.457	3.421	3.489
Time (13) [ms]	4.039	3.888	3.684	3.687	3.668
Time (15) [ms]	4.056	3.862	3.713	3.655	3.712

Table 12: _THREADS_PER_BLOCK 1024, Grid-Stride Loop Comparison

3-2-1. _THREADS_PER_BLOCK = 256, Throughput

Kernel	Original	Shared Memory	Grid-stride Loop (8)
Memory Throughput 1 [%]	84.67	84.93	87.68
Memory Throughput 3 [%]	85.63	85.56	88.50
Memory Throughput 5 [%]	86.42	87.06	86.85
Memory Throughput 7 [%]	85.56	86.43	86.25
Memory Throughput 9 [%]	85.05	86.58	86.57
Memory Throughput 11 [%]	88.67	84.85	87.36
Memory Throughput 13 [%]	83.73	87.34	84.35
Memory Throughput 15 [%]	76.76	87.29	83.71

Table 13: _THREADS_PER_BLOCK 256, _GRID_REDUCTION_FACTOR 8, Memory Throughput

Kernel	Original	Shared Memory	Grid-stride Loop (8)
Compute Throughput 1 [%]	15.36	17.29	14.84
Compute Throughput 3 [%]	26.14	26.49	15.01
Compute Throughput 5 [%]	30.56	27.88	35.21
Compute Throughput 7 [%]	42.80	32.74	37.03
Compute Throughput 9 [%]	47.09	36.27	57.70
Compute Throughput 11 [%]	58.72	43.25	56.09
Compute Throughput 13 [%]	61.37	46.09	76.74
Compute Throughput 15 [%]	66.84	51.48	75.86

Table 14: _THREADS_PER_BLOCK 256, _GRID_REDUCTION_FACTOR 8, Compute Throughput

3-2-2. _THREADS_PER_BLOCK = 1024, Throughput

Kernel	Original	Shared Memory	Grid-stride Loop (16)
Memory Throughput 1 [%]	90.23	85.10	84.56
Memory Throughput 3 [%]	86.12	82.05	87.40
Memory Throughput 5 [%]	84.58	80.10	90.43
Memory Throughput 7 [%]	80.36	76.97	87.07
Memory Throughput 9 [%]	77.28	73.76	86.56
Memory Throughput 11 [%]	69.44	70.71	85.14
Memory Throughput 13 [%]	65.67	70.82	76.83
Memory Throughput 15 [%]	59.34	67.47	76.20

Table 15: _THREADS_PER_BLOCK 1024, _GRID_REDUCTION_FACTOR 16, Memory Throughput

Kernel	Original	Shared Memory	Grid-stride Loop (16)
Compute Throughput 1 [%]	15.62	17.17	14.28

Compute Throughput 3 [%]	26.19	22.99	14.71
Compute Throughput 5 [%]	30.44	24.05	36.97
Compute Throughput 7 [%]	39.54	29.10	37.78
Compute Throughput 9 [%]	43.25	30.14	55.00
Compute Throughput 11 [%]	47.74	33.34	56.63
Compute Throughput 13 [%]	48.64	36.67	67.45
Compute Throughput 15 [%]	51.31	38.71	65.95

Table 16: _THREADS_PER_BLOCK 1024, _GRID_REDUCTION_FACTOR 16, Compute Throughput

3-2-3. Achieved Occupancy

_THREADS_PER_BLOCK	64	128	256	512	1024
Original [%]	86.98	86.42	84.93	83.44	61.79
Shared Memory [%]	85.38	92.47	90.82	91.14	63.59
Grid-Stride Loop [%]	96.06	96.38	95.42	94.19	63.63

Table 16: Achieved Occupancy by _THREADS_PER_BLOCK

3-2-4. L1/TEX Cache Hit Rate

_THREADS_PER_BLOCK	64	128	256	512	1024
Original [%]	89.00	89.10	89.42	89.58	89.66
Shared Memory [%]	5.64	0.63	0.01	0.00	0.00
Grid-Stride Loop [%]	88.72	89.09	89.42	89.58	89.66

Table 17: L1/TEX Cache Hit Rate by _THREADS_PER_BLOCK

3-2-5. L2 Cache Hit Rate

_THREADS_PER_BLOCK	64	128	256	512	1024
Original [%]	56.65	55.99	57.49	54.74	54.33
Shared Memory [%]	55.57	59.83	51.07	57.51	54.30
Grid-Stride Loop [%]	54.04	52.45	50.85	58.36	49.91

Table 18: L2 Cache Hit Rate by _THREADS_PER_BLOCK

3-3. 스크린샷

```

[STRIDE(8)] CUDA kernel launch with 524288(4194385) blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.742(ms) / speedup = 292.883
Test PASSED

[STRIDE(16)] CUDA kernel launch with 262144(4194385) blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.595(ms) / speedup = 304.882
Test PASSED

[STRIDE(32)] CUDA kernel launch with 131072(4194385) blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.653(ms) / speedup = 300.040
Test PASSED

== 1D Gaussian filtering of image of 268435459 pixels(KERNEL WIDTH = 15) ==

[HOST] Time to filter an image of 268435459 pixels on the host = 1384.846(ms)

[ORIGINAL] CUDA kernel launch with 4194385 blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.587(ms) / speedup = 386.118
Test PASSED

[SHARED] CUDA kernel launch with 4194385 blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.547(ms) / speedup = 390.463
Test PASSED

[STRIDE(2)] CUDA kernel launch with 2097152(4194385) blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.527(ms) / speedup = 392.684
Test PASSED

[STRIDE(4)] CUDA kernel launch with 1048576(4194385) blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.499(ms) / speedup = 395.781
Test PASSED

[STRIDE(8)] CUDA kernel launch with 524288(4194385) blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.612(ms) / speedup = 383.416
Test PASSED

[STRIDE(16)] CUDA kernel launch with 262144(4194385) blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.757(ms) / speedup = 368.558
Test PASSED

[STRIDE(32)] CUDA kernel launch with 131072(4194385) blocks of 64 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.823(ms) / speedup = 362.263
Test PASSED

^^ Done

-----
C:\HW_2_Gaussian_Filtering_1D_Three_Methods\x64\Release\Gaussian_Filtering_1D_Three_Methods.exe(프로세스 17000)이 (가) 0 코드(0x0)와 함께 종료되었습니다.
이 창을 보려면 맥부 기어 누르세요...

```

```

[STRIDE(8)] CUDA kernel launch with 262144(2097153) blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.638(ms) / speedup = 298.793
Test PASSED

[STRIDE(16)] CUDA kernel launch with 131072(2097153) blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.677(ms) / speedup = 295.603
Test PASSED

[STRIDE(32)] CUDA kernel launch with 65536(2097153) blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.983(ms) / speedup = 272.877
Test PASSED

== 1D Gaussian filtering of image of 268435459 pixels(KERNEL WIDTH = 15) ==

[HOST] Time to filter an image of 268435459 pixels on the host = 1345.660(ms)

[ORIGINAL] CUDA kernel launch with 2097153 blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.740(ms) / speedup = 359.798
Test PASSED

[SHARED] CUDA kernel launch with 2097153 blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.679(ms) / speedup = 365.765
Test PASSED

[STRIDE(2)] CUDA kernel launch with 1048576(2097153) blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.668(ms) / speedup = 366.907
Test PASSED

[STRIDE(4)] CUDA kernel launch with 524288(2097153) blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.676(ms) / speedup = 366.049
Test PASSED

[STRIDE(8)] CUDA kernel launch with 262144(2097153) blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.715(ms) / speedup = 362.175
Test PASSED

[STRIDE(16)] CUDA kernel launch with 131072(2097153) blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.613(ms) / speedup = 372.487
Test PASSED

[STRIDE(32)] CUDA kernel launch with 65536(2097153) blocks of 128 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.900(ms) / speedup = 345.059
Test PASSED

^^ Done

-----
C:\HW_2_Gaussian_Filtering_1D_Three_Methods\x64\Release\Gaussian_Filtering_1D_Three_Methods.exe(프로세스 26164)이 (가) 0 코드(0x0)와 함께 종료되었습니다.
이 창을 보려면 맥부 기어 누르세요...

```

```
[STRIDE(8)] CUDA kernel launch with 131072(1048577) blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.495(ms) / speedup = 307.073
Test PASSED

[STRIDE(16)] CUDA kernel launch with 65536(1048577) blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.548(ms) / speedup = 302.444
Test PASSED

[STRIDE(32)] CUDA kernel launch with 32768(1048577) blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.661(ms) / speedup = 293.150
Test PASSED

=== 1D Gaussian filtering of image of 268435459 pixels(KERNEL WIDTH = 15) ===

[HOST] Time to filter an image of 268435459 pixels on the host = 1392.491(ms)

[ORIGINAL] CUDA kernel launch with 1048577 blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.569(ms) / speedup = 390.155
Test PASSED

[SHARED] CUDA kernel launch with 1048577 blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.573(ms) / speedup = 389.677
Test PASSED

[STRIDE(2)] CUDA kernel launch with 524288(1048577) blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.458(ms) / speedup = 402.643
Test PASSED

[STRIDE(4)] CUDA kernel launch with 262144(1048577) blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.521(ms) / speedup = 395.503
Test PASSED

[STRIDE(8)] CUDA kernel launch with 131072(1048577) blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.543(ms) / speedup = 393.011
Test PASSED

[STRIDE(16)] CUDA kernel launch with 65536(1048577) blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.557(ms) / speedup = 391.434
Test PASSED

[STRIDE(32)] CUDA kernel launch with 32768(1048577) blocks of 256 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.700(ms) / speedup = 376.299
Test PASSED

*** Done

C:\HW_2_Gaussian_Filtering_1D_Three_Methods\x64\Release\Gaussian_Filtering_1D_Three_Methods.exe(프로세스 11044)이(가) 0 코드(0x0)와 함께 종료되었습니다.
이 창을 닫으려면 아무 키나 누르세요...
```

```
[STRIDE(8)] CUDA kernel launch with 65536(524289) blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.486(ms) / speedup = 306.842
Test PASSED

[STRIDE(16)] CUDA kernel launch with 32768(524289) blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.528(ms) / speedup = 303.178
Test PASSED

[STRIDE(32)] CUDA kernel launch with 16384(524289) blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.646(ms) / speedup = 293.331
Test PASSED

=== 1D Gaussian filtering of image of 268435459 pixels(KERNEL WIDTH = 15) ===

[HOST] Time to filter an image of 268435459 pixels on the host = 1371.023(ms)

[ORIGINAL] CUDA kernel launch with 524289 blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.660(ms) / speedup = 374.862
Test PASSED

[SHARED] CUDA kernel launch with 524289 blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.553(ms) / speedup = 386.153
Test PASSED

[STRIDE(2)] CUDA kernel launch with 262144(524289) blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.497(ms) / speedup = 392.342
Test PASSED

[STRIDE(4)] CUDA kernel launch with 131072(524289) blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.407(ms) / speedup = 393.407
Test PASSED

[STRIDE(8)] CUDA kernel launch with 65536(524289) blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.481(ms) / speedup = 394.060
Test PASSED

[STRIDE(16)] CUDA kernel launch with 32768(524289) blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.542(ms) / speedup = 387.286
Test PASSED

[STRIDE(32)] CUDA kernel launch with 16384(524289) blocks of 512 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.642(ms) / speedup = 376.635
Test PASSED

*** Done

C:\HW_2_Gaussian_Filtering_1D_Three_Methods\x64\Release\Gaussian_Filtering_1D_Three_Methods.exe(프로세스 18152)이(가) 0 코드(0x0)와 함께 종료되었습니다.
이 창을 닫으려면 아무 키나 누르세요...
```

```
[STRIDE(8)] CUDA kernel launch with 32768(262145) blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.684(ms) / speedup = 302.304
Test PASSED

[STRIDE(16)] CUDA kernel launch with 16384(262145) blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.687(ms) / speedup = 302.068
Test PASSED

[STRIDE(32)] CUDA kernel launch with 8192(262145) blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.668(ms) / speedup = 303.631
Test PASSED

=== 1D Gaussian filtering of image of 268435459 pixels(KERNEL WIDTH = 15) ===

[HOST] Time to filter an image of 268435459 pixels on the host = 1363.692(ms)

[ORIGINAL] CUDA kernel launch with 262145 blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 4.401(ms) / speedup = 309.894
Test PASSED

[SHARED] CUDA kernel launch with 262145 blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.999(ms) / speedup = 340.972
Test PASSED

[STRIDE(2)] CUDA kernel launch with 131072(262145) blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 4.056(ms) / speedup = 336.251
Test PASSED

[STRIDE(4)] CUDA kernel launch with 65536(262145) blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.862(ms) / speedup = 353.077
Test PASSED

[STRIDE(8)] CUDA kernel launch with 32768(262145) blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.713(ms) / speedup = 367.230
Test PASSED

[STRIDE(16)] CUDA kernel launch with 16384(262145) blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.655(ms) / speedup = 373.091
Test PASSED

[STRIDE(32)] CUDA kernel launch with 8192(262145) blocks of 1024 threads
Average Time to filter an image of 268435459 pixels on the device (32 tests) = 3.712(ms) / speedup = 367.343
Test PASSED

*** Done

C:\Program Files\NVIDIA Corporation\NVIDIA GeForce RTX 4080 SUPER\Release\Gaussian_Filtering_1D_Three_Methods.exe(프로세스 29960)(가) 0 코드(0x0)와 함께 종료되었습니다.
이 창을 닫으려면 아무 키나 누르세요...
```

Documents

block64.ncvrep X

Result

629- apply_1D_Gaussian_wrap_device_ORIGINAL

Size

(4194305, 1, 1)(x64, 1, 1)

Time

3.55 ms

Cycles

8,141,025

GPU

0 - NVIDIA GeForce RTX 4080 SUPER

SM Frequency

2.29 GHz

Process

[19960] Gaussian_Filtering_1D_Three_Methods.exe

Attributes

Current

Summary

Details

Source

Context

Comments

Raw

Session

Compare

Tools

View

Export

This table shows all results in the report. Use the column headers to sort the results in this report. Double-click a result to see detailed metrics. Double-click on demangled names to rename it.

ID	Estimated Speedup [x]	Function Name	Demangled Name	Duration Runtime Improvement [ms]	Compute Throughput	Memory Throughput	# Registers	Register Grid Size	Block Size [block]	Result Type	
0	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.55	0.00	14.25	86.90	23	4194305, 1, ...	64, 1, ...	Kernel SIMT
1	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.57	0.00	14.04	83.70	23	4194305, 1, ...	64, 1, ...	Kernel SIMT
2	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.56	0.00	17.05	83.90	22	4194305, 1, ...	64, 1, ...	Kernel SIMT
3	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	4.38	0.00	17.92	83.71	22	4194305, 1, ...	64, 1, ...	Kernel SIMT
4	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.95	0.00	17.09	84.22	28	2097153, 1, ...	64, 1, ...	Kernel SIMT
5	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.35	0.00	17.02	86.07	28	2097153, 1, ...	64, 1, ...	Kernel SIMT
6	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.24	0.00	15.20	85.90	26	1948576, 1, ...	64, 1, ...	Kernel SIMT
7	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.52	0.00	16.22	84.73	28	1948576, 1, ...	64, 1, ...	Kernel SIMT
8	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.49	0.00	14.14	85.00	28	524288, 1, ...	64, 1, ...	Kernel SIMT
9	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.50	0.00	15.60	85.01	28	524288, 1, ...	64, 1, ...	Kernel SIMT
10	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.41	0.00	14.00	86.90	28	262144, 1, ...	64, 1, ...	Kernel SIMT
11	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.40	0.00	15.13	87.15	28	262144, 1, ...	64, 1, ...	Kernel SIMT
12	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.46	0.00	13.46	85.87	26	131072, 1, ...	64, 1, ...	Kernel SIMT
13	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.53	0.00	13.91	80.11	28	131072, 1, ...	64, 1, ...	Kernel SIMT
14	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	4.06	0.00	27.20	84.85	23	4194305, 1, ...	64, 1, ...	Kernel SIMT
15	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.45	0.00	27.11	86.13	23	4194305, 1, ...	64, 1, ...	Kernel SIMT
16	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.94	0.00	29.50	86.37	22	4194305, 1, ...	64, 1, ...	Kernel SIMT
17	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.42	0.00	28.69	86.59	22	4194305, 1, ...	64, 1, ...	Kernel SIMT
18	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.40	0.00	17.45	86.00	28	2097153, 1, ...	64, 1, ...	Kernel SIMT
19	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.50	0.00	16.04	86.84	28	2097153, 1, ...	64, 1, ...	Kernel SIMT
20	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.53	0.00	16.77	86.83	28	1948576, 1, ...	64, 1, ...	Kernel SIMT
21	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.40	0.00	15.70	87.42	28	1948576, 1, ...	64, 1, ...	Kernel SIMT

block128-nvcc.exe

Result

629- apply_1D_Gaussian_wrap_device_ORIGINAL

Size

(2097153, 1, 1)(x128, 1, 1)

Time

3.41 ms

Cycles

7,815,221

GPU

0 - NVIDIA GeForce RTX 4080 SUPER

SM Frequency

2.29 GHz

Process

[33044] Gaussian_Filtering_1D_Three_Methods.exe

Attributes

Current

Summary

Details

Source

Context

Comments

Raw

Session

Compare

Tools

View

Export

This table shows all results in the report. Use the column headers to sort the results in this report. Double-click a result to see detailed metrics. Double-click on demangled names to rename it.

ID	Estimated Speedup [x]	Function Name	Demangled Name	Duration Runtime Improvement [ms]	Compute Throughput	Memory Throughput	# Registers	Register Grid Size	Block Size [block]	Result Type	
0	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.41	0.00	14.40	86.87	23	2097153, 1, ...	128, 1, ...	Kernel SIMT
1	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.51	0.00	14.63	85.80	23	2097153, 1, ...	128, 1, ...	Kernel SIMT
2	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.52	0.00	17.19	85.37	22	2097153, 1, ...	128, 1, ...	Kernel SIMT
3	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.33	0.00	17.07	88.19	22	2097153, 1, ...	128, 1, ...	Kernel SIMT
4	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.07	0.00	17.18	85.79	28	1948576, 1, ...	128, 1, ...	Kernel SIMT
5	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.42	0.00	16.55	86.87	28	1948576, 1, ...	128, 1, ...	Kernel SIMT
6	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.39	0.00	15.07	87.14	28	524288, 1, ...	128, 1, ...	Kernel SIMT
7	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.36	0.00	15.18	87.69	28	524288, 1, ...	128, 1, ...	Kernel SIMT
8	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.36	0.00	15.32	88.03	28	262144, 1, ...	128, 1, ...	Kernel SIMT
9	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.99	0.00	15.61	85.15	28	262144, 1, ...	128, 1, ...	Kernel SIMT
10	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.38	0.00	13.83	87.38	28	131072, 1, ...	128, 1, ...	Kernel SIMT
11	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.50	0.00	14.60	85.00	28	131072, 1, ...	128, 1, ...	Kernel SIMT
12	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	4.50	0.00	13.37	81.71	28	65536, 1, ...	128, 1, ...	Kernel SIMT
13	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	4.49	0.00	13.99	82.83	28	65536, 1, ...	128, 1, ...	Kernel SIMT
14	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.53	0.00	27.59	84.99	23	2097153, 1, ...	128, 1, ...	Kernel SIMT
15	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.47	0.00	26.23	85.86	23	2097153, 1, ...	128, 1, ...	Kernel SIMT
16	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.48	0.00	27.37	85.90	22	2097153, 1, ...	128, 1, ...	Kernel SIMT
17	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	4.00	0.00	26.81	84.92	22	2097153, 1, ...	128, 1, ...	Kernel SIMT
18	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.52	0.00	16.49	86.81	28	1948576, 1, ...	128, 1, ...	Kernel SIMT
19	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.36	0.00	17.18	86.48	28	1948576, 1, ...	128, 1, ...	Kernel SIMT
20	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.42	0.00	16.20	86.70	28	524288, 1, ...	128, 1, ...	Kernel SIMT
21	0.00	apply_1D_Gaussian...	apply_1D_Gaussian...	3.41	0.00	16.19	86.89	28	524288, 1, ...	128, 1, ...	Kernel SIMT

Documents

block256.ncu-rep X

Result: 629 - apply_1D_Gaussian_wrap_device_ORIGINAL Size: (1048577, 1, 1)x(256, 1, 1) Time: 3.34 ms Cycles: 7,660,911 GPU: 0 - NVIDIA GeForce RTX 4080 SUPER SM Frequency: 2.29 GHz Process: [B260] Gaussian_Filtering_1D_Three_Methods.exe Attributes: Compare Tools View Export

Summary Details Source Context Comments Raw Session

This table shows all results in the report. Use the column headers to sort the results in this report. Double-click a result to see detailed metrics. Double-click on demangled names to rename it.

ID	Estimated Speedup [x]	Function Name	Demangled Name	Duration Runtime Improvement [ms]	Compute Throughput	Memory Throughput [# Registers [register/t] Grid Size	Block Size [block]	Result Type
0	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.34	0.00	13.44 88.86 23 1848577, 1, 1	256, 1, 1	Kernel SIMT
1	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.33	0.00	15.36 88.87 23 1848577, 1, 1	256, 1, 1	Kernel SIMT
2	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.34	0.00	17.94 88.84 22 1848577, 1, 1	256, 1, 1	Kernel SIMT
3	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.32	0.00	17.29 88.93 22 1848577, 1, 1	256, 1, 1	Kernel SIMT
4	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.39	0.00	17.77 87.84 28 524288, 1, 1	256, 1, 1	Kernel SIMT
5	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.46	0.00	17.16 85.92 28 524288, 1, 1	256, 1, 1	Kernel SIMT
6	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.43	0.00	16.18 88.84 28 524288, 1, 1	256, 1, 1	Kernel SIMT
7	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.31	0.00	16.29 85.86 28 524288, 1, 1	256, 1, 1	Kernel SIMT
8	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.43	0.00	14.44 86.56 28 131072, 1, 1	256, 1, 1	Kernel SIMT
9	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.37	0.00	14.84 87.68 28 131072, 1, 1	256, 1, 1	Kernel SIMT
10	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.37	0.00	14.86 83.65 28 65536, 1, 1	256, 1, 1	Kernel SIMT
11	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.37	0.00	13.99 87.53 28 65536, 1, 1	256, 1, 1	Kernel SIMT
12	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.43	0.00	14.13 86.37 28 32768, 1, 1	256, 1, 1	Kernel SIMT
13	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.37	0.00	13.93 85.81 28 32768, 1, 1	256, 1, 1	Kernel SIMT
14	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.36	0.00	27.76 85.88 23 1848577, 1, 1	256, 1, 1	Kernel SIMT
15	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.40	0.00	26.14 85.63 23 1848577, 1, 1	256, 1, 1	Kernel SIMT
16	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.39	0.00	25.47 87.34 22 1848577, 1, 1	256, 1, 1	Kernel SIMT
17	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.40	0.00	26.49 85.56 22 1848577, 1, 1	256, 1, 1	Kernel SIMT
18	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.37	0.00	17.69 85.66 28 524288, 1, 1	256, 1, 1	Kernel SIMT
19	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.46	0.00	17.07 86.78 28 524288, 1, 1	256, 1, 1	Kernel SIMT
20	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.30	0.00	16.28 87.37 28 524288, 1, 1	256, 1, 1	Kernel SIMT
21	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.44	0.00	16.35 86.52 28 524288, 1, 1	256, 1, 1	Kernel SIMT

Documents

block512.ncu-rep X

Result: 629 - apply_1D_Gaussian_wrap_device_ORIGINAL Size: (524289, 1, 1)x(512, 1, 1) Time: 3.44 ms Cycles: 7,895,306 GPU: 0 - NVIDIA GeForce RTX 4080 SUPER SM Frequency: 2.29 GHz Process: [2138] Gaussian_Filtering_1D_Three_Methods.exe Attributes: Compare Tools View Export

Summary Details Source Context Comments Raw Session

This table shows all results in the report. Use the column headers to sort the results in this report. Double-click a result to see detailed metrics. Double-click on demangled names to rename it.

ID	Estimated Speedup [x]	Function Name	Demangled Name	Duration Runtime Improvement [ms]	Compute Throughput	Memory Throughput [# Registers [register/t] Grid Size	Block Size [block]	Result Type
0	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.54	0.00	15.20 86.35 23 524289, 1, 1	512, 1, 1	Kernel SIMT
1	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.52	0.00	14.59 88.70 23 524289, 1, 1	512, 1, 1	Kernel SIMT
2	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.40	0.00	18.00 86.96 22 524289, 1, 1	512, 1, 1	Kernel SIMT
3	0.00	apply_1D_Gaussian	apply_1D_Gaussian	4.15	0.00	17.20 85.85 22 524289, 1, 1	512, 1, 1	Kernel SIMT
4	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.82	0.00	17.27 87.97 28 262148, 1, 1	512, 1, 1	Kernel SIMT
5	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.49	0.00	17.29 86.60 28 262148, 1, 1	512, 1, 1	Kernel SIMT
6	0.00	apply_1D_Gaussian	apply_1D_Gaussian	4.68	0.00	16.29 84.77 28 131072, 1, 1	512, 1, 1	Kernel SIMT
7	0.00	apply_1D_Gaussian	apply_1D_Gaussian	4.46	0.00	15.47 83.44 28 131072, 1, 1	512, 1, 1	Kernel SIMT
8	0.00	apply_1D_Gaussian	apply_1D_Gaussian	4.35	0.00	15.65 85.42 28 65536, 1, 1	512, 1, 1	Kernel SIMT
9	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.35	0.00	14.89 87.73 28 65536, 1, 1	512, 1, 1	Kernel SIMT
10	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.37	0.00	14.98 88.21 28 32768, 1, 1	512, 1, 1	Kernel SIMT
11	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.54	0.00	14.33 88.60 28 32768, 1, 1	512, 1, 1	Kernel SIMT
12	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.59	0.00	13.64 85.30 28 16384, 1, 1	512, 1, 1	Kernel SIMT
13	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.56	0.00	14.84 88.25 28 16384, 1, 1	512, 1, 1	Kernel SIMT
14	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.54	0.00	26.52 85.34 23 524289, 1, 1	512, 1, 1	Kernel SIMT
15	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.28	0.00	26.21 89.77 23 524289, 1, 1	512, 1, 1	Kernel SIMT
16	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.36	0.00	25.53 88.44 22 524289, 1, 1	512, 1, 1	Kernel SIMT
17	0.00	apply_1D_Gaussian	apply_1D_Gaussian	4.33	0.00	24.60 89.07 22 524289, 1, 1	512, 1, 1	Kernel SIMT
18	0.00	apply_1D_Gaussian	apply_1D_Gaussian	4.19	0.00	17.52 86.86 28 262148, 1, 1	512, 1, 1	Kernel SIMT
19	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.37	0.00	17.69 87.36 28 262148, 1, 1	512, 1, 1	Kernel SIMT
20	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.49	0.00	16.31 85.28 28 131072, 1, 1	512, 1, 1	Kernel SIMT
21	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.80	0.00	16.44 87.82 28 131072, 1, 1	512, 1, 1	Kernel SIMT

(Recaptured version for full metrics, values may vary)

Documents

block1024.ncu-rep X

Result: 629 - apply_1D_Gaussian_wrap_device_ORIGINAL Size: (262145, 1, 1)x(1024, 1, 1) Time: 3.25 ms Cycles: 7,478,138 GPU: 0 - NVIDIA GeForce RTX 4080 SUPER SM Frequency: 2.29 GHz Process: [19748] Gaussian_Filtering_1D_Three_Methods.exe Attributes: Compare Tools View Export

Summary Details Source Context Comments Raw Session

This table shows all results in the report. Use the column headers to sort the results in this report. Double-click a result to see detailed metrics. Double-click on demangled names to rename it.

ID	Estimated Speedup [x]	Function Name	Demangled Name	Duration Runtime Improvement [ms]	Compute Throughput	Memory Throughput [# Registers [register/t] Grid Size	Block Size [block]	Result Type
0	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.25	0.00	15.79 98.41 23 262145, 1, 1	1024, 1, 1	Kernel SIMT
1	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.26	0.00	15.79 98.41 23 262145, 1, 1	1024, 1, 1	Kernel SIMT
2	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.45	0.00	17.56 88.49 22 262145, 1, 1	1024, 1, 1	Kernel SIMT
3	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.37	0.00	17.45 86.49 22 262145, 1, 1	1024, 1, 1	Kernel SIMT
4	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.49	0.00	17.97 87.19 28 131072, 1, 1	1024, 1, 1	Kernel SIMT
5	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.48	0.00	17.55 85.79 28 131072, 1, 1	1024, 1, 1	Kernel SIMT
6	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.45	0.00	15.99 85.77 28 65536, 1, 1	1024, 1, 1	Kernel SIMT
7	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.49	0.00	16.20 86.58 28 65536, 1, 1	1024, 1, 1	Kernel SIMT
8	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.41	0.00	15.38 86.75 28 32768, 1, 1	1024, 1, 1	Kernel SIMT
9	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.46	0.00	15.99 85.83 28 32768, 1, 1	1024, 1, 1	Kernel SIMT
10	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.39	0.00	16.95 86.84 28 16384, 1, 1	1024, 1, 1	Kernel SIMT
11	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.41	0.00	14.83 86.48 28 16384, 1, 1	1024, 1, 1	Kernel SIMT
12	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.53	0.00	14.04 84.18 28 8192, 1, 1	1024, 1, 1	Kernel SIMT
13	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.54	0.00	14.10 84.84 28 8192, 1, 1	1024, 1, 1	Kernel SIMT
14	11.31	apply_1D_Gaussian	apply_1D_Gaussian	3.42	0.39	27.88 86.53 23 262145, 1, 1	1024, 1, 1	Kernel SIMT
15	11.03	apply_1D_Gaussian	apply_1D_Gaussian	3.41	0.43	26.57 85.89 23 262145, 1, 1	1024, 1, 1	Kernel SIMT
16	9.00	apply_1D_Gaussian	apply_1D_Gaussian	3.57	0.46	23.67 82.89 22 262145, 1, 1	1024, 1, 1	Kernel SIMT
17	0.00	apply_1D_Gaussian	apply_1D_Gaussian	3.59	0.00	23.34 82.85 22 262145, 1, 1	1024, 1, 1	Kernel SIMT
18	11.48	apply_1D_Gaussian	apply_1D_Gaussian	3.33	0.38	18.38 98.21 28 131072, 1, 1	1024, 1, 1	Kernel SIMT
19	10.08	apply_1D_Gaussian	apply_1D_Gaussian	3.37	0.36	18.16 88.61 28 131072, 1, 1	1024, 1, 1	Kernel SIMT
20	11.28	apply_1D_Gaussian	apply_1D_Gaussian	3.35	0.37	16.47 88.17 28 65536, 1, 1	1024, 1, 1	Kernel SIMT
21	11.05	apply_1D_Gaussian	apply_1D_Gaussian	3.41	0.38	16.15 86.85 28 65536, 1, 1	1024, 1, 1	Kernel SIMT